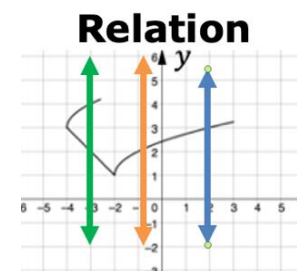


10.4 Is It a Function?

Lesson Introduction

	Benchmark	MA.8.F.1.1 Given a set of ordered pairs, a table, a graph, or mapping diagram, determine whether the relationship is a function. Identify the domain and range of the relation.			Learning Target	I can determine whether a relation is a function.
	Benchmark Coherence	Building On			Working Toward	
		N/A			MA.912.F.1.2 Given a function represented in function notation, evaluate the function for an input in its domain. For a real-world context, interpret the output.	
	Essential Questions	- How can you use a mapping diagram to determine whether a relation is a function? - How can you use the vertical line test to determine if the graph of a relation represents a function? - How can you determine if a set of ordered pairs represents a function?				
	Lesson Narrative	Students determine if relations represented in a set of ordered pairs, a table, a graph, or mapping diagram are functions. They explore how the vertical line test can be used to determine if a relation represented by a graph is a function. They also analyze samples of students' work for errors.				
	Component Summary	10.4.1 Warm-Up (~5 min.)	Watch and reflect on a video about an engineering student's perseverance (<i>SE p. 135</i>)	10.4.4 Your Turn! (5-10 min.)	Demonstrate knowledge of functions by analyzing arguments, completing diagrams, and recognizing what is not a function (<i>SE p. 139</i>)	
		10.4.2 Guided Practice (10-15 min.)	Explain if given relations are functions (<i>SE p. 136</i>)	10.4.5 Wrap-Up (~5 min.)	Analyze student arguments in determining if a relation represented in a table is a function (<i>SE p. 140</i>)	
		10.4.3 Exploration (10 min.)	Explore how the vertical line test is used to determine if a relation represented by a graph is a function (<i>SE p. 138</i>)			
	Resources	Glossary		Materials		Additional Preparation
		<u>Key Terms:</u> - Vertical line test <u>Additional Terms:</u> - Function		(Optional) Ruler, markers, and chart paper		- The 10.4.1 Warm-Up is accompanied by a video to play for students (1:30 minutes). (Optional) Sort rulers, markers, and chart paper by group for student-created anchor charts displaying the vertical line test.

 Teacher Tips	Common Misconceptions - Students may believe that two of the same input with the same unique output could not represent a function. For example, students may interpret the relation $\{(2,3), (4,5), (2,3)\}$ as not a function, even though the input of 2 still has only one unique output. - Students may mistakenly conduct a horizontal line test instead of a vertical line test to check if a relation is a function. - Students may think “the range” refers to the input or x-values and “the domain” refers to the output or y-values.	Things to Consider Building MA.K12.MTR.2.1: Multiple Representations - Throughout the lesson, students are asked to determine if relations represented in a set of ordered pairs, a table, a graph, or mapping diagram are functions. Students must know how to identify the domain and range in each representation. Then they need to check if each element in the domain corresponds to exactly one element in the range in each representation.
	Literacy and Language Routines Clarify, Critique, Correct Consider prompting students to utilize this instructional routine for 10.4.4 Your Turn!. By scaffolding their responses, students are not only given the opportunity to refine their responses for more precise answers but also make space for building confidence as they collaborate with a classmate for positive feedback.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection Students are prompted to analyze the thinking and reasoning of other students in writing (10.4.2 Guided Practice, 10.4.4 Your Turn!, and 10.4.5 Wrap-Up). Notice the use of sentence frames in 10.4.2 Guided Practice. Consider prompting students to use this scaffolding for their practice in 10.4.4 Your Turn! or orienting students to each other by having them read each explanation in 10.4.4 Your Turn! out loud and responding verbally before refining their responses in writing. Students can then independently apply these strategies during 10.4.5 Wrap-Up.
	 Differentiate Instruction	Vertical Line Test: 10.4.3 Exploration provides an opportunity for students to construct their own entry points for making sense of what defines a relation as a function. - Provide chart paper and markers for students to craft their own working definitions and examples. - Use your classroom floor as real-life chart paper, pasting a variety of relations or functions on the ground for students to walk the (vertical) line to determine whether each is a function. Strategies should be paired with meaningful questions for students to build conceptual understanding. - Does it matter which line we use to determine whether a relation is a function? - What if we only drew the blue line? Would this relation then be a function? What about the green line? - What does that make you think about when determining whether a relation is a function on a graph?
Lesson Notes:		



10.4.1 Warm-Up: Sticking With It

Component Overview: Students watch a motivational video (1:30 minutes) about an engineering student's perseverance and reflect on when they have stuck with something that was hard in order to reach a goal.



10.4.1 Warm-Up: Sticking With It

Engineering students learn many lessons both in and out of the classroom. One student shared what she learned: "Sometimes there is a lot of pressure that can cause unwanted strain, but it is what happens at the point of failure that reveals the amount of strength that is really inside."

1. What is one life lesson you have learned this school year?
Answers will vary. Possible answers could include learning how to handle stress and time management.
2. When is a time when you have had to stick with something that was hard in order to reach a goal?
Answers will vary. Possible answers could include struggling with a concept in class or saving money for an item.



Do you know someone who is an engineer, computer scientist, or programmer? Consider hosting a guest speaker panel to answer students' questions about attending college and/or STEM careers.

10.4.2 Guided Practice: Is It a Function?

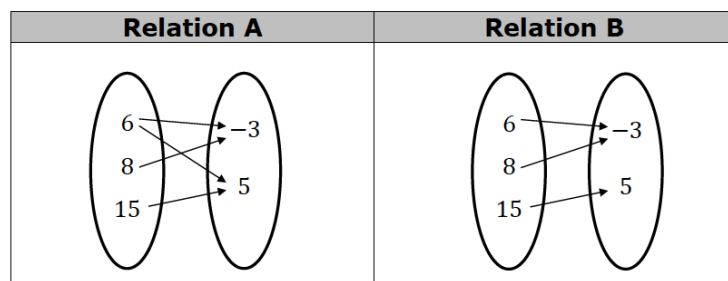
Component Overview: Students practice determining if relations represented in a set of ordered pairs, a table, a graph, or mapping diagram are functions. Consider allowing students to create and use reference cards outlining how to check if relations are functions.



10.4.2 Guided Practice: Is It a Function?

A function is a mathematical relation for which each element in the domain corresponds to exactly one element in the range.

1. Two relations are shown below.



- Mila said that neither Relation A nor Relation B is a function.
- K'Drien said that Relation A is not a function, but reasoned that Relation B is a function.

Complete the statements.

I agree with

- ☐ Mila.
☒ K'Drien.

I know that Relation B

- ☒ is
☐ is not

a function because

each element in the

- ☒ domain
☐ range

- ☒ does
☐ does not

correspond with exactly one element in the

- ☐ domain.
☒ range.



Students may interpret Relation B as “not a function” because both inputs 6 and 8 share an output of -3 . Clarify this misconception by prompting students to define what it means to be a function using Relation B as an example.



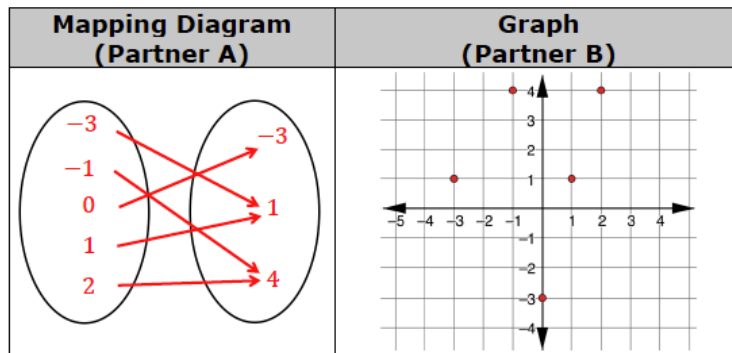
Consider an inquiry-based approach when identifying relations as functions from a mapping diagram. What similarities and differences do you see between the two mapping diagrams? What do you look for when determining whether Relation A or B is a function?

10.4.2 Guided Practice: Is It a Function? (continued)

2. A relation is given as a set of ordered pairs.

$$\{(2, 4), (1, 1), (-1, 4), (-3, 1), (0, -3)\}$$

- a. Determine with your partner who will be Partner A and who will be Partner B. Represent the relation using the method associated with your letter.



- b. Check your partner's representation. Once you both agree that each representation is correct, complete the other representation in your book.
- c. Complete the statements to explain how each representation shows whether the relation is a function.

The **mapping diagram** has

- ☐ only one arrow to
 - ☐ **only one arrow from**
 - ☐ more than one arrow to
 - ☐ more than one arrow from
- each element in the

- ☐ **domain**
 - ☐ **range**
- to its corresponding element in the

- ☐ domain,
 - ☐ **range,**
- therefore the relation

- ☐ **is**
- ☐ is not

a function.



Encourage students to write the ordered pairs in order by the x -values before creating the mapping diagram. While this is not required, it may positively contribute to procedural fluency.



If necessary, consider pausing to highlight connections between ordered pairs and their representations on a graph.

How can you tell if a relation represented as a set of ordered pairs is a function?

How can you tell if a relation represented as a graph is a function?



MA.K12.MTR.2.1: Multiple Representations

Throughout the lesson students are asked to determine if relations represented in a set of ordered pairs, a table, a graph, or mapping diagram are functions. Students must know how to identify the domain and range from each representation. Then they need to check if each element in the domain corresponds to exactly one element in the range in each representation.

10.4.2 Guided Practice: Is It a Function? (continued)

The **graph** demonstrates the relation ☒ **is** ☐ **is not** a

function because ☒ **each x -value** ☐ **at least one x -value**

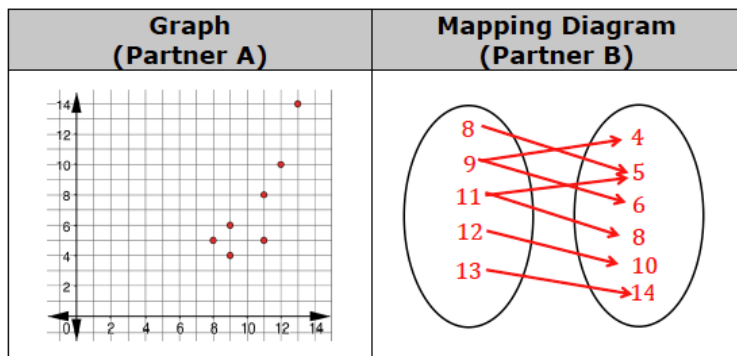
corresponds to ☒ **exactly one y -value** ☐ **more than one y -value**

on the graph.

3. The table below gives the number of letters of the winning word of the Scripps National Spelling Bee, and the number of people killed by venomous spiders for the years 1996-2006.

	1999	2000	2001	2002	2003	2004	2005	2006
Letters in winning word	9	8	11	12	11	13	12	9
People killed by venomous spider	6	5	5	10	8	14	10	4

- a. Using the same partner letters as in the previous problem, represent the relation with a mapping diagram and as a graph.



If necessary, consider pausing to highlight connections between input and output in the table of values. How can you tell if a relation represented as a table is a function?

If an input value repeats and is paired with the same output, will the relation be a function?

What do you notice is different about the graph when the relation is not a function?



Consider providing highlighters or colored pencils for students to make use of such tools when identifying inputs and outputs.

10.4.2 Guided Practice: Is It a Function? (continued)

- b. Check your partner's representation. Once you both agree that each representation is correct, then complete the other representation.
- c. Use the definition of a function to justify whether the relationship between the number of letters of the winning word of the Scripps National Spelling Bee and the number of people killed by venomous spiders is a function or not.

The relation is not a function because two elements in the domain, 9 and 11, both correspond to more than one element in the range.



Students may need scaffolding to help using the definition from Lesson 3. A function is a mathematical relation for which each element in the domain corresponds to exactly one element in the range. Use these questions to help students who need scaffolding:

- What are key words in the definition?
- What do you know about a relation?
- What are the elements in the domain?
- What are the elements in the range?

10.4.3 Exploration: Vertical Line Test

Component Overview: Students explore how the vertical line test can be used to determine if a relation represented by a graph is a function. Students will also apply their findings to determine whether relations are functions from a graph and ordered pairs.



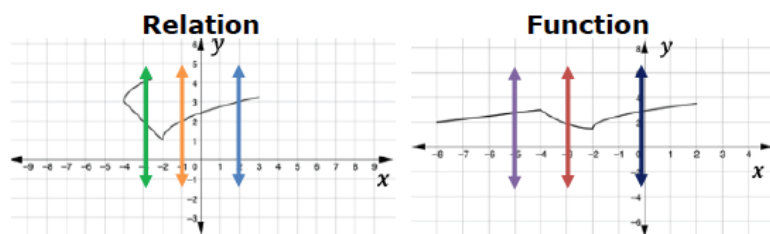
10.4.3 Exploration: Vertical Line Test



The vertical line test can be used to determine if a relation represented by a graph is a function.

The vertical lines drawn on the graph of a function show whether each x -value corresponds to exactly one y -value.

A graph of a relation and a function are shown with three different vertical lines drawn on each.



- Complete the explanation that describes how the vertical line test can be used to show the relation represented by a graph is not a function.

The
☐ blue
☐ orange
☒ green
 line shows that the relation is not a

function because the line demonstrates that for the

x -value of
☒ -3
☐ -1
☐ 2
 there is more than one y -value.



Position students as sense-makers during this Exploration, making space for productive struggle as well as student-crafted responses and representations.

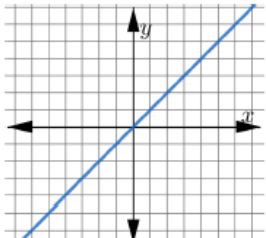
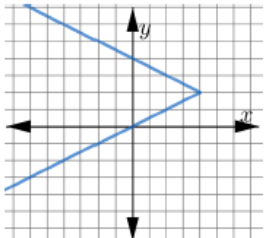
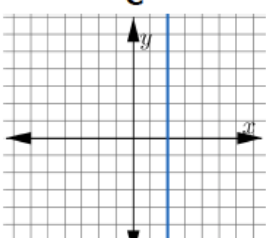
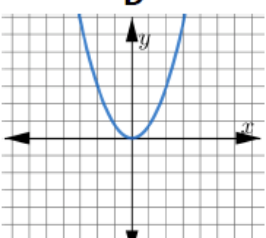


The vertical line test activates many entry points for learners. Using sketched lines provides visual scaffolding, while using a pencil and a finger activates kinesthetic connections for students as they discover what the intersecting lines mean for the function. Consider prompting students to create their own anchor chart during the Exploration. Encourage creativity and collective learning (MA.K12.MTR.1.1) as well as multiple representations (MA.K12.MTR.2.1).

10.4.3 Exploration: Vertical Line Test (continued)

2. Sort the relations into each category by writing the letters that label the relations in the table.

Functions	Not Functions
<i>A, D, F, G</i>	<i>B, C, E, H</i>

A 	B 
C 	D 
E $\{(2, 7), (3, 7), (2, -1), (7, 3)\}$	F $\{(-5, -1), (9, 3), (4, -1), (-2, 6)\}$
G $\{(1, 9), (-2, 8), (0, 12), (-6, 0)\}$	H $\{(-5, 8), (6, -4), (2, 11), (-5, -7)\}$



Consider prompting questions for students who could use additional support during the Exploration.

Graph C: What do you notice about the vertical line test and Graph C? Does this mean it passes or does not pass? Provide an example using coordinate pairs.

Graph D: How is Graph D similar to Graph B? How is it different? What does that make you think about regarding whether or not each is a function?

Relation E: What do you notice about the x -values? What if both y -values for the same x -value were the same?

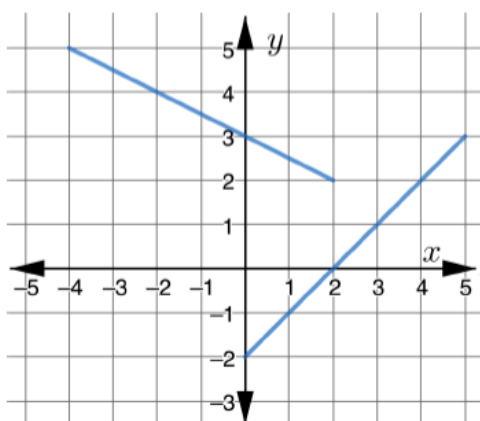
10.4.4 Your Turn!

Component Overview: Students analyze samples of student work for errors and determine if relations represented in graphs and mapping diagrams are functions. Students determine if input and output values can be mapped to show if a relation is a function. Also, students identify sets of ordered pairs that are not functions.



10.4.4 Your Turn!

1. A group of students were discussing whether the relationship on the graph is a function.



Ed: The relationship is not a function because if I draw a horizontal line at $y = 2$, the horizontal line crosses the graph at two points.

Debra: The relationship is a function because it is a graph of two different lines.

Alejandro: The relationship is not a function because $x = 1$ has two different y -values.

Which students are **not** correct and what is the flaw in their argument?

Ed is incorrect because he needs to use a vertical line not a horizontal line.

Debra is incorrect because the graph is one relation not two different lines. The relation fails the vertical line test.



Clarify, Critique, and Correct

Consider framing this activity as a routine that gives students opportunities to critique the reasoning of others. Consider having students present their findings to a partner, refine their responses based on their partner, and share how their responses changed.



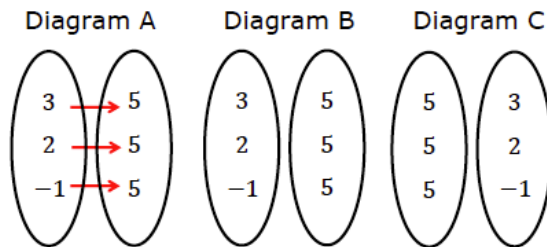
Note that students could share that Alejandro is partially correct, as $x = 1$ is only one example of an input that shares more than one output.



Challenge students to write an inequality that represents the values (or domain) that makes it not a function.

10.4.4 Your Turn! (continued)

2. Consider the following incomplete mapping diagrams.



a. Complete Diagram A so that it is a function.

Answers will vary.

b. Is it possible to complete Diagram B so that it is not a function? If so, complete the diagram to show a relation, but not a function. If not, justify your reasoning.

Answers will vary. No, because there is only one output possible in Diagram B, so every input will only correspond to one output.

c. Is it possible to complete the mapping diagram for Diagram C so it represents a function? If so, complete the diagram to show a function. If not, justify your reasoning.

Answers will vary. No, because there is only one input possible in Diagram B so it will have three output values.

3. Select all the relations that are **not** functions.

- ☐ $\{(1, 3), (3, 7), (5, 11), (7, 15), (9, 19)\}$
- ☒ $\{(1, 3), (1, 7), (5, 11), (5, 15), (9, 19)\}$
- ☐ $\{(-2, 4), (-1, 1), (0, 0), (1, 1), (2, 4)\}$
- ☒ $\{(2, 4), (1, 1), (0, 0), (1, -1), (2, -4)\}$
- ☐ $\{(6, 3), (4, 1), (2, 1), (0, -1), (-2, -3)\}$
- ☒ $\{(1, 3), (3, 7), (3, 11), (7, 15), (9, 19)\}$
- ☐ $\{(1, 3), (3, 7), (5, 11), (9, 15), (9, 19)\}$



Clarify, Critique, and Correct (continued)
Students can continue refining their responses with this instructional routine for questions 2a-c.



Extend student thinking by prompting them to edit the relations that are not functions to make them functions.



If pressed for time, question 3 can be saved for homework.

10.4.5 Wrap-Up: Error Analysis

Component Overview: Students are given a table of values and two samples of students' reasoning for determining if a relation is a function. Students analyze the two samples to determine if there are errors and flaws in their reasoning.



10.4.5 Wrap-Up: Error Analysis

The table shows the gold-medal winners of the 100-meter dash in the Olympic games since 1980.

Winner	Height (meters)	Time (seconds)
Allan Wells (GBR)	1.83	10.25
Carl Lewis (USA)	1.88	9.99
Carl Lewis (USA)	1.88	9.92
Linford Christie (GBR)	1.87	9.96
Donovan Bailey (CAN)	1.85	9.84
Maurice Greene (USA)	1.76	9.87
Justin Gatlin (USA)	1.86	9.85
Usain Bolt (JAM)	1.95	9.69
Usain Bolt (JAM)	1.95	9.63
Usain Bolt (JAM)	1.95	9.81

Both Olivia and Liam wrote an explanation about whether the table of values represents a function.

Olivia's Explanation	Liam's Explanation
The table of values of the gold-medal winners is a function because the winners competed in different years.	The table of values of the gold-medal winners is a function. Although there are some inputs that give different outputs, those inputs are for the same person.



Incorporate an opportunity to build literacy by prompting students to respond to both Olivia and Liam by writing an explanation in response to each of their explanations.

1. Complete the statements to evaluate both explanations.

I agree with

- ☐ Liam.
- ☐ Olivia.
- ☒ neither Liam nor Olivia.

- ☐ Liam did not use
- ☐ Olivia did not use
- ☒ Neither Liam nor Olivia used

the correct

- ☒ domain
- ☐ range

to determine if the relation in the table is a function.